SPE-228929-MS

Pushing Ultrasonic Logging Limits: First Cement Evaluation in 26-Inch Od, 1.25-Inch Wall Casing

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

The growing complexity of offshore well designs has led to the use of increasingly large conductor casings above 22 inches in diameter with wall thicknesses greater than 1 inch. While these configurations meet demanding structural requirements, they present major challenges for cement evaluation. Conventional CBL-VDL sonic tools are ineffective in such environments, while ultrasonic measurements are heavily affected by signal attenuation and limited energy transmission through thick steel. These limitations become particularly critical in offshore wells where confirmation of cement presence is required before drilling can proceed.

This paper presents a case where a 26-inch conductor was set 120 m shallower than planned, with no visible cement returns at the seabed. Without confirmation of cement around the shoe, drilling and blowout preventer (BOP) installation could not continue, creating a significant operational risk and potential cost escalation.

A new cement evaluation approach was implemented to overcome these limitations. The solution combined low-frequency, high–signal-to-noise-ratio ultrasonic transducers, a large rotating logging head to minimize standoff effects, custom-engineered centralizers for optimal tool positioning, and enhanced data processing algorithms that reduced turnaround time from 12–24 hours to just four hours. **This approach delivered the first-ever successful cement evaluation in a 26-inch OD, 1.25-inch-thick casing —confirming cement presence behind the casing and around the shoe.** This verification enabled safe BOP installation and continuation of drilling without the need for costly re-spudding or abandonment.

This achievement extends the operational limits of cement evaluation in extreme casing environments, improves decision-making in high-risk offshore scenarios, reduces rig time, and lowers the environmental footprint of drilling operations.

Introduction

Well integrity assurance and zonal isolation are critical for both new and mature wells. In offshore environments, any compromise in cement evaluation can lead to severe operational, environmental, and financial consequences. Cement evaluation is essential to verify the effectiveness of well construction, prevent potential failures, and ensure compliance with regulatory standards.

The growing complexity of offshore well designs has led to the use of large-diameter conductor casings above 22 inches in diameter with wall thicknesses greater than 1 inch. While these configurations meet demanding structural requirements, they also present major challenges for cement evaluation. In many cases, cement verification relies on observing returns at the seabed, often confirmed via bull's-eye with a remotely operated vehicle (ROV) (Figure 1). When returns are absent, the cement must be verified before drilling can continue; otherwise, the well would need to be re-spudded. In these large-diameter conductor applications, cement is required not only to provide zonal isolation but also to deliver the mechanical support necessary for safe blowout preventer (BOP) installation. Prior to the work described in this paper, logging in such casing sizes and thicknesses was not considered technically feasible, leaving operators with limited options for cement verification.

Figure 1—Cement returns verification at the seafloor (adapted from "Deepwater Cementing Best Practices for the Riserless Section," C. Gill, Nexen Petroleum; G.A. Fuller; R. Faul; 2005).

Large, heavy-wall conductors create conditions that push conventional evaluation methods to their limits. CBL-VDL sonic tools are ineffective in such environments, and even ultrasonic technology—used commonly for well integrity assessment—loses effectiveness in casings above 13 ⅜ in. OD, wall thickness over 0.7 in., or in high-density drilling fluids. Under these conditions, ultrasonic signals suffer high attenuation, resulting in poor data quality and making accurate cement evaluation extremely difficult.

To overcome these limitations, a novel approach was developed to extend the operating range of ultrasonic cement and casing measurement tools beyond their conventional capabilities. This solution combined next-generation low-frequency, high–signal-to-noise-ratio transducers capable of exciting resonance in >1 in.-thick casings, a large rotating logging head to minimize standoff, a high-torque motor section for rotating large subs, and custom-engineered centralizers to maintain optimal tool positioning. In addition, an extended search-window processing algorithm was implemented to improve pulse-echo reliability, significantly reducing data turnaround time from 12–24 hours to just four hours.

The result was the first-ever successful cement evaluation in a 26-inch OD, 1.25-inch-thick conductor casing. This paper presents that case study in detail, exploring the challenges of evaluating cement in extreme casing environments, the technical innovations that enabled success, and the impact on operational efficiency. The findings demonstrate how next-generation ultrasonic technology can deliver high-quality cement evaluation logs in conditions previously considered beyond reach, ensuring well integrity while reducing rig time, operational costs, and carbon emissions.

History Of Cement Evaluation Tools

Cement evaluation tools have evolved significantly, driven by the need for more accurate and reliable assessments of well integrity. Early methods relied on sonic tools, followed by ultrasonic and flexural wave-based technologies, each incorporating improvements in measurement accuracy, resolution, and interpretation capability.

Early Sonic-Based CBL-VDL Measurements

The earliest cement evaluation tools were sonic-based systems with omnidirectional transmitters and receivers. Cement bonding was assessed by analyzing the first-arrival amplitude attenuation, a method known as the cement bond log (CBL). The later addition of a second receiver, typically spaced 5 ft from the transmitter, enabled acquisition of the variable density log (VDL), providing qualitative visualization of casing-to-cement and cement-to-formation bonding through full waveform analysis.

A standard CBL-VDL tool configuration includes a transmitter, a 3-ft spaced receiver for CBL amplitude measurement, and a 5-ft spaced receiver for VDL waveform acquisition (Figure 2). Receiver spacing may vary by service provider. Due to the omnidirectional nature of the transmitter and receivers, these tools do not provide azimuthal information. High first-arrival amplitude generally indicates poor cement bonding (free pipe), while low amplitude suggests a strong cement bond.

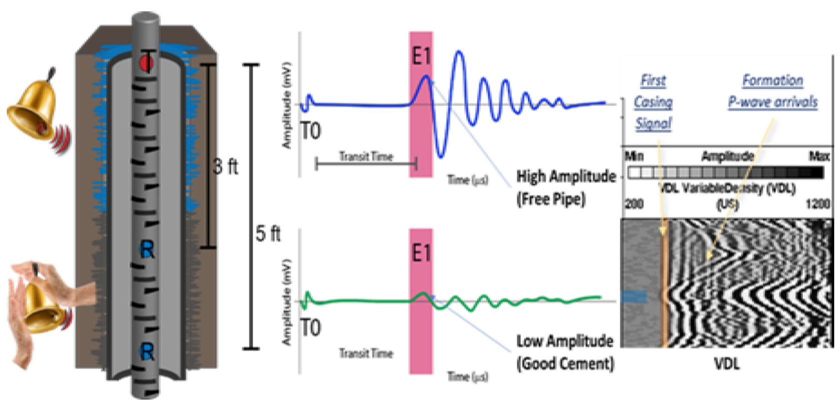

Figure 2—CBL-VDL tool measurement. Two waveforms showing poor bond response (top) and good bond response (bottom).

VDL data enhances interpretation by offering qualitative insight into bonding through analysis of the full sonic waveform, particularly useful for evaluating cement-to-formation bonding. Despite decades of technological advancements, the CBL-VDL remains a robust and widely used method, either as a standalone technique or in combination with newer tools. Its core principle remains unchanged: evaluating cement bond integrity by analyzing acoustic energy attenuation in the first-arrival signal.

However, conventional sonic tools have limitations in detecting cement channeling, lightweight slurries, or cement in the presence of a wet microannulus. To address these shortcomings, CBL tools were enhanced with directional receivers arranged in arrays of six or eight azimuthal elements. This configuration allows low-resolution azimuthal cement evaluation (45° radial resolution) and improves detection of nonuniform bonding compared with traditional omnidirectional CBL-VDL tools.

Advancements in Ultrasonic Cement Evaluation Tools

In parallel with CBL-VDL developments, ultrasonic cement evaluation tools evolved. The second-generation ultrasonic tools, introduced in the early 1980s, incorporated multiple radially distributed ultrasonic sensors to provide azimuthal cement quality assessment as well as casing radius and thickness measurements.

Subsequent advancements led to the development of rotating ultrasonic tools with a single transducer mounted on a rotating sub, delivering full 360° azimuthal cement evaluation. These tools remain in use today, offering high-resolution coverage, while pad-mounted acoustic measurement tools—equipped with six azimuthally spaced pads—were also introduced for specific applications.

Ultrasonic Pulse-Echo Measurement

Ultrasonic pulse-echo measurement determines cement impedance, casing radius, and wall thickness. In this method, a transducer mounted on a rotating sub emits an ultrasonic pulse that travels through the wellbore fluid, reflects off the casing wall, and returns to the transducer. Multiple azimuthal measurements (typically 36, 72, or up to 180) provide comprehensive coverage. However, large casing sizes and heavy drilling muds can weaken reflections, reducing signal quality.

The acoustic impedance of the material behind the casing is calculated from the decay rate of resonance reflections (Figure 3). Low acoustic impedance indicates the presence of liquid or gas, while high impedance corresponds to solid materials such as cement or formation rock. Acoustic impedance (Z) is defined by Equation (1), as:

$$Z = \rho \cdot v \qquad \text{Equation 1}$$

where ρ is material density and ν is acoustic velocity, expressed in (MRayl). Table 1 shows typical acoustic impedance values for common wellbore and annular materials.

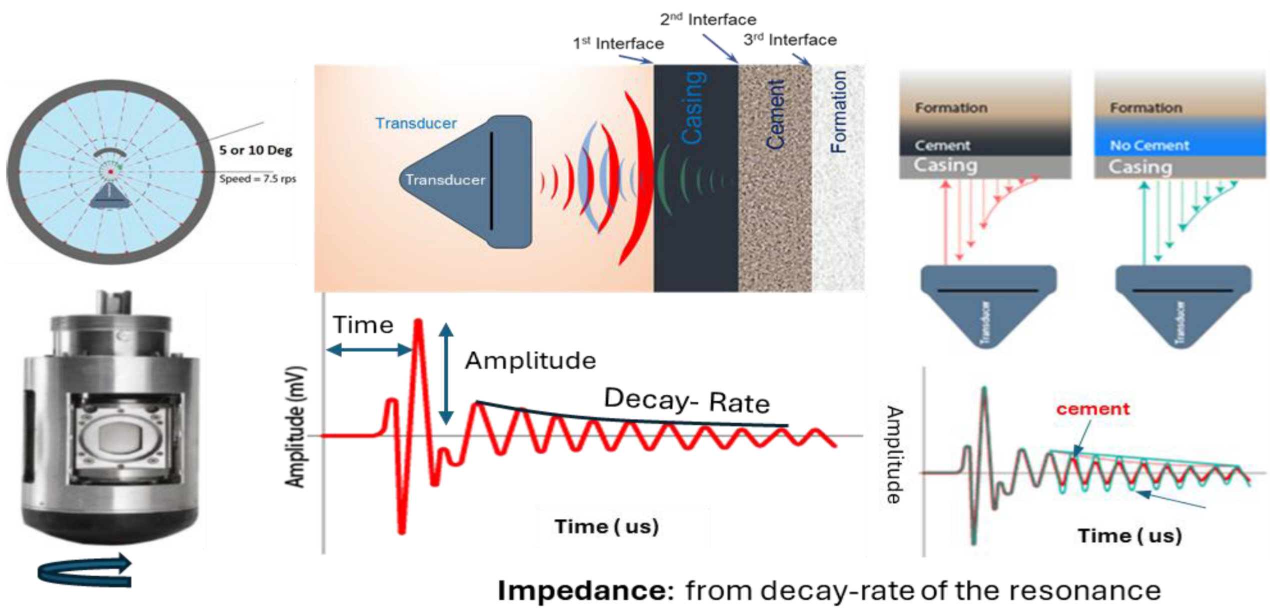

Figure 3—Traditional pulse echo ultrasonic measurement, illustrating the rotating sub and waveform response in time

Table 1—Acoustic Impedance of Different Materials

Substance	Density (kg/m³)	Acoustic Impedance (MRayls)
Gas	< 350	<0.1
Water (25°C)	997	1.5
Crude Oil	910	1.3
Diesel	803	1
Drilling fluids	1,000-2,000	1.5-3
Cement (light)	1,400	3.1-3.6
Cement (Class G)	1,900	5-7

The combination of ultrasonic and CBL-VDL measurements is the preferred approach for comprehensive cement evaluation. Rotating ultrasonic tools provide high-resolution azimuthal cement images, casing radius, and wall thickness measurements—with azimuthal resolution as fine as 2° and vertical resolution of approximately 0.6 inches—enabling improved detection of liquid channels behind casing. CBL-VDL measurements complement ultrasonics by providing independent bond assessment, mitigating limitations associated with either technology when used alone (AlArfaj et al., 2022).

While these technologies have significantly advanced cement evaluation capabilities, their effectiveness diminishes in extreme casing environments. In particular, very large-diameter, heavy-wall conductors present conditions where both CBL-VDL and ultrasonic tools encounter fundamental performance limitations. These specific challenges and their impact on cement evaluation are discussed in the following section.

Challenges in cement evaluation for large-diameter, heavy-wall casings

Ultrasonic cement evaluation in very large-diameter, thick-walled casings presents a number of technical challenges that limit data quality and reliability. These limitations affect the accuracy of cement bond assessment and well integrity evaluation in extreme casing environments.

Key challenges include:

- **Ultrasonic Signal Attenuation and Standoff Distance** – Large casing diameters increase the standoff distance between the tool and the casing wall, causing significant signal attenuation. This leads to weak or even missing return signals during ultrasonic logging, reducing measurement reliability.
- **Frequency Limitations of Standard Transducers** – Conventional ultrasonic transducers operate in the 250–700 kHz range, suitable for casings up to 0.6 inches thick. Thick-wall casings resonate at lower frequencies (around 100 kHz), where standard transducers are less effective, resulting in degraded pulse-echo measurement quality.
- **Centralization Challenges** – Accurate cement evaluation depends on effective tool centralization. In large casings, achieving proper centralization is difficult, as most commercially available centralizers are limited to around 20 inches OD. Poor centralization increases standoff variation and introduces measurement inaccuracies.
- **Field Acquisition Software Limitations** – Current field acquisition software is not enabled to handle pulse-echo data from casings thicker than 1.0 inch or larger than 24 inches OD, due to the absence of 3D correction tables. As a result, such data must be processed offline using MATLAB engineering code, which significantly extends the processing time and result delivery.

In addition, **CBL-VDL measurements** face further challenges in large-diameter casings. Centralization remains an issue, and the amplitude contrast between free pipe and bonded cement is minimal, increasing the risk of misinterpretation. For example, in a 20-inch, 126 ppf casing cemented with a 12.5 ppg slurry (acoustic impedance ≈ 3.8 MRayl):

- Free pipe CBL amplitude ≈ 35 mV
- 80% bonded cement CBL amplitude ≈ 27 mV

The 8 mV contrast is insufficient for reliable interpretation, especially when eccentricity further distorts readings. This makes it difficult to confidently distinguish between unbonded and bonded cement in such environments.

Extending the range of cement evaluation

The increasing use of thick, large-diameter casings—often deployed with heavy mud systems—has created well conditions that exceed the operating envelope of traditional ultrasonic cement evaluation tools. Legacy tools, typically limited to 13 3/8-inch OD and wall thickness up to 0.6 inches, operate in the 250–700 kHz range. In heavy-wall casings, this frequency range is ineffective due to insufficient low-frequency energy and greater acoustic attenuation, leading to poor data quality.

To address these limitations, several key innovations were deployed, collectively extending the range and capability of ultrasonic cement evaluation:

- **Powerful New-Generation Transducers** – These transducers deliver up to a 30-fold improvement in signal-to-noise ratio, with enhanced energy output in the 100–300 kHz range. The increased low-frequency energy improves penetration through thick steel, enabling reliable measurements in casings greater than 1 inch thick (Figure 4; Thierry et al., 2016).
- **Large Rotating Logging Heads** – Larger rotating heads reduce the standoff distance between the transducer and the casing (Figure 5), improving signal quality and stability (Thierry et al., 2016).
- **Optimized Centralization** – A new large-hole centralizer enables effective tool positioning in 14–22 inch OD casings (Figure 6a). For casings above 22 inches, custom-made centralizers were developed to maintain proper tool centering (Figure 6b), minimizing measurement errors caused by eccentricity.
- **Enhanced Data Processing Workflow** –Previously, evaluating thick casing greater than 1 in. required post-job processing using MATLAB code, significantly delaying result delivery. The expansion of the thickness search-window algorithm allows direct processing of thick-casing data without MATLAB post-processing. This change reduced turnaround time from 12–24 hours to just 4 hours.

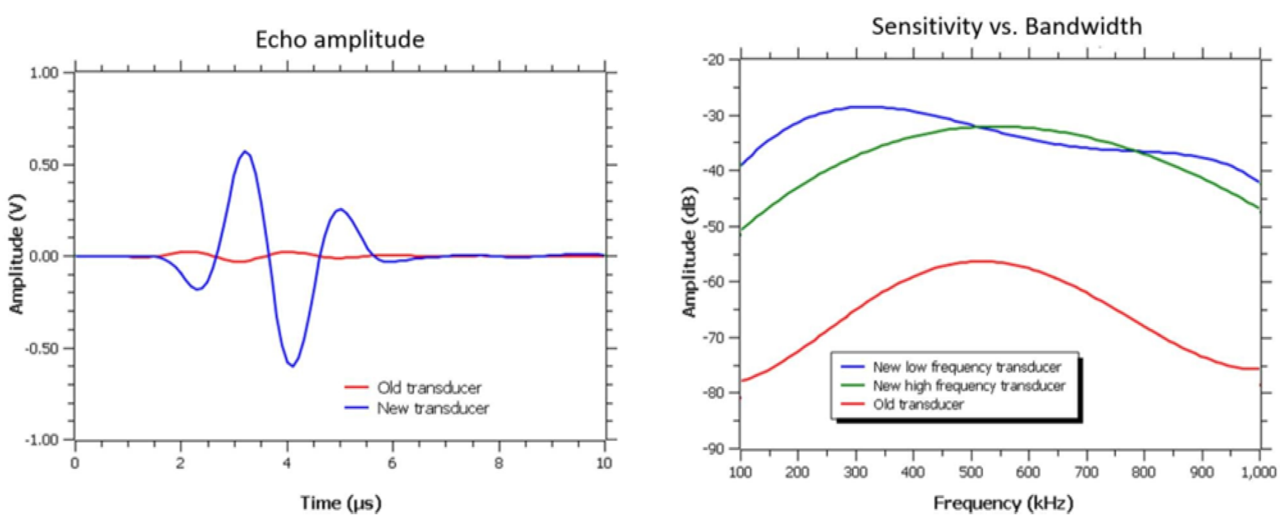

Figure 4—Comparison of echoes acquired using legacy and new-generation ultrasonic transducers under identical measurement conditions. Right: Comparison showing at least a 30 dB increase in signal response for the new-generation transducer relative to the legacy design.

Figure 5—New large logging heads for pulse-echo measurements for 26-in. casing.

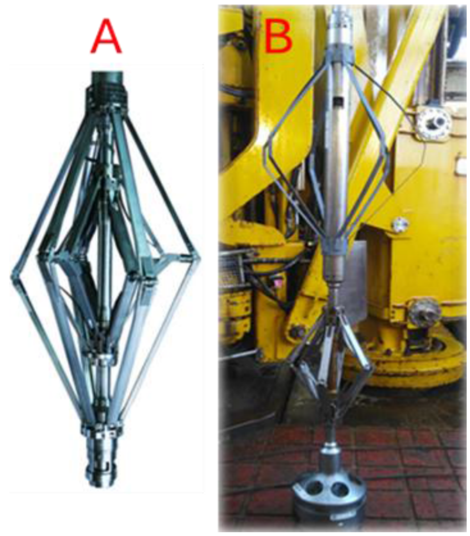

Figure 6—A) Newly developed large-hole centralizer enabling tool centralization in 14–22 in. OD casings. (B) Custom locally manufactured centralizers for 26 in. OD casing.

By enhancing signal strength, improving centralization, and streamlining data processing, these hardware and software advancements have expanded the operational limits of ultrasonic cement evaluation. Tools can now reliably inspect casings larger than 13 3/8 inches OD and thicker than 0.6 inches, even in heavy mud systems where acoustic attenuation is severe (Figure 7).

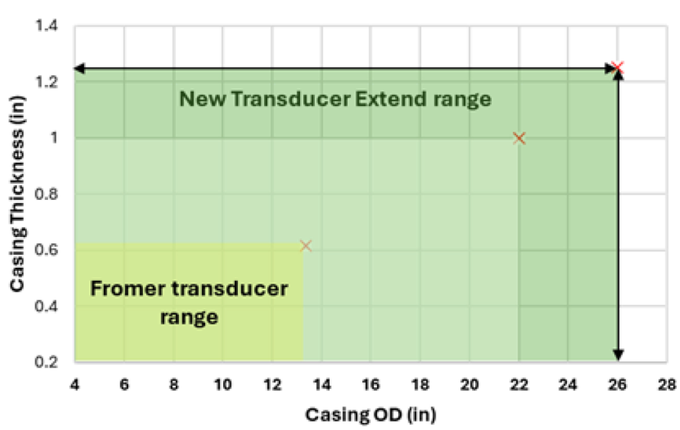

Figure 7—Range of casing dimensions for former and new transducer.

Case study: cement evaluation of a 26-inch conductor casing

The hardware, centralization, and data processing innovations described in the previous section were applied in the field for the world's **first successful cement evaluation of a 26-inch OD, 1.25-inch wall conductor casing**. This case presented an ideal opportunity to validate the extended capabilities of the upgraded ultrasonic measurement under the most demanding conditions.

Background

In this case, a cement evaluation log was required for a 26-inch conductor casing set 120 m shallower than planned. The cementing objective was to place cement to the seabed; however, during placement, returns at the seabed were not confirmed by the ROV, raising concerns about whether the conductor had adequate support for blowout preventer (BOP) installation.

To meet construction requirements and ensure structural integrity, a cement evaluation was performed to determine the **cement height in the annulus** before proceeding to the next drilling phase.

Without verified cement presence and height, drilling could not proceed and the well risked a costly re-spud.

The 26-inch OD casing had a weight of 330.4 lbm/ft and a nominal wall thickness of 1.25 inches. It was cemented using a 13 ppg lead slurry followed by a 16 ppg Class G tail slurry, with 200% excess volume pumped to maximize the likelihood of full cement coverage to the seabed. Despite this, no cement returns were confirmed by the ROV during the cementing operation. Final well details are shown in Figure 8.

Well Data Summary

Item	Data	Planned/Actual
Well Depth:	406 m	Planned
Shoe Depth:	289 m	**Actual**
Cement Top:	211 m	Planned
Lead Density	1.58 SG	**Actual**
Top of Lead	211m	Planned
Bottom of Lead	249 m	Planned
Tail Density	1.92 SG	**Actual**
Top of Lead	249m	Planned
Bottom of Lead	289 m	Planned
Mud Line Depth:	211 m	Planned
Excess:	200%	

Hole

Dia.	OH Mean Ann. Excess.	OH Mean Dia.	From	To
36.000 in	200 %	50.359 in	211 m	406 m

Casing

OD	ID	Weight	From	To
26.000 in	23.500 in	330.29 lbm/ft	0 m	289 m

Figure 8—Well parameters and conductor casing details.

Cement Evaluation Logging Methodology

To verify cement height and determine the top of cement (TOC), a pulse-echo ultrasonic tool was deployed in combination with a CBL-VDL. The ultrasonic system featured:

- **Large rotating logging head** (11.4-inch OD) driven by a high-torque motor to ensure stable rotation.
- **Next-generation high-power transducer** designed to deliver strong, high-quality signals in the low-frequency range required for accurate measurements in thick-walled casings.
- **Custom centralizers** designed to fit the 23.5-inch casing ID and maintain eccentricity within 0.4 inches.

The CBL-VDL was centralized using a Powered 4-arm caliper to acquire complementary amplitude and waveform data for correlation and verification of the TOC.

While corrosion measurements (radius and wall thickness) were also recorded and clearly demonstrating the casing condition, detailed corrosion analysis was not performed, as the primary objective was to confirm cement placement and structural support for BOP installation.

Log Review and Interpretation

The ultrasonic log response across the 26-inch casing was of high quality. Tool centralization remained within tolerance, measured wall thickness and radius matched nominal values, and weld signatures were clearly identified across amplitude, radius, thickness, and acoustic impedance logs—confirming measurement integrity.

Based on the log responses in (Figure 9), the results can be divided into two sections:

- **Section 1 – Cemented Interval (285–335 m)**
  Below the TOC at 235 m MD, the pulse-echo acoustic impedance log indicated a well-cemented section corresponding to the 16 ppg Class G tail slurry. Measured thickness averaged 1.22 inches, radius 11.75 inches, and acoustic impedance ~6.8 MRayl, consistent with fully set Class G cement (1.92 g/cm^3). The CBL showed only a minor amplitude reduction (average 24 mV) compared with free pipe, limiting its reliability for TOC determination.
- **Section 2 – Free Pipe Interval (above TOC at 235 m MD)**
  The pulse-echo measurement indicated a liquid-filled annulus with an average acoustic impedance of ~1.5 MRayl. Thickness and radius matched those in the cemented section. The VDL displayed clear pipe ringing, and CBL amplitudes averaged 28 mV—slightly higher than the cemented interval but with insufficient contrast for a confident TOC pick.

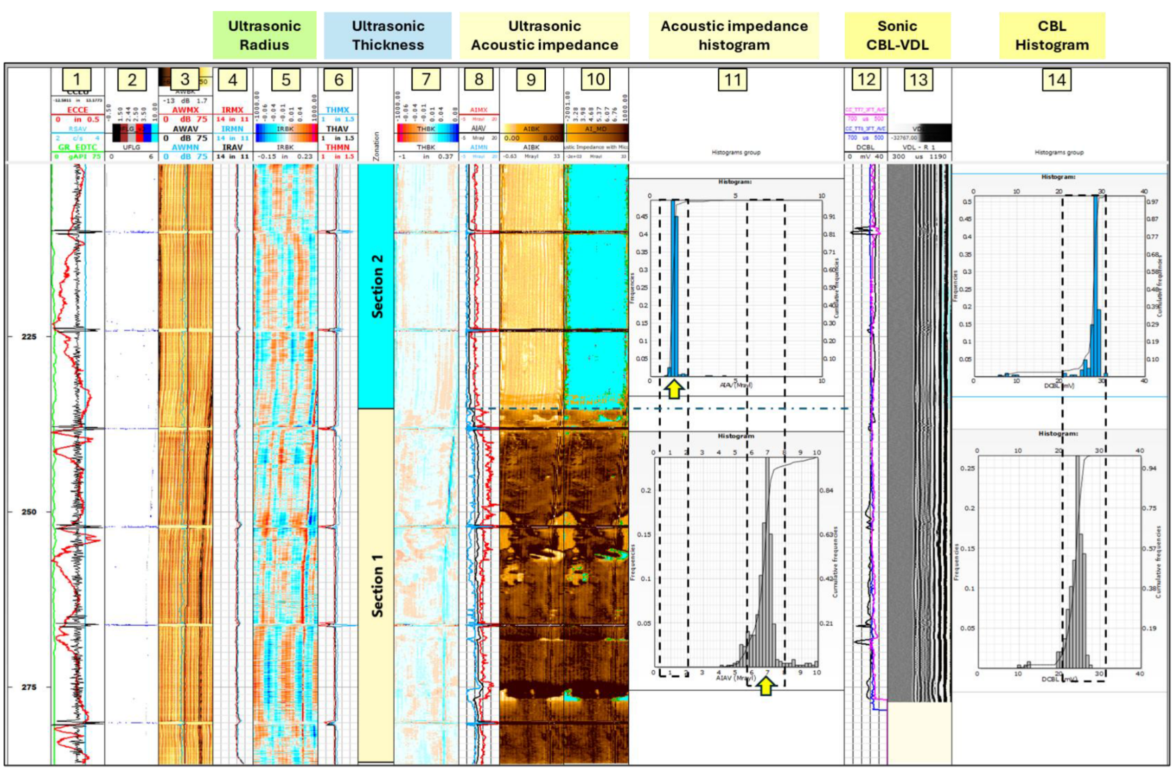

Figure 9—Pulse-echo measurement log combined with CBL-VDL response.

Key Findings and Logging Comparison

Analysis of the acoustic impedance histogram (Track 11, Figure 9) shows that the pulse-echo ultrasonic measurement provided a clear and unambiguous identification of the TOC and cemented sections. There was a distinct separation between the acoustic impedance values of cement and those of liquid-filled intervals, enabling confident placement of the TOC on the log.

In comparison, the CBL histogram (Track 14, Figure 9) lacked clear separation. Due to the limited contrast between free pipe and bonded cement, CBL and VDL logs alone are insufficient for making a reliable TOC determination in very large and thick wall casing, reinforcing the limitations of CBL/VDL in large-diameter, thick-wall casings.

In contrast, the CBL-VDL response (Track 13, Figure 9) did not provide a reliable indication of the TOC. The minimal amplitude difference between free pipe (~28 mV) and bonded cement (~24 mV) in this large, thick-walled casing made it impossible to distinguish between cemented and uncemented sections with confidence. This limitation was further compounded by tool centralization challenges and the inherently low acoustic energy penetration of sonic measurements in heavy-wall conductor.

In summary, for this case the pulse-echo ultrasonic method was the only measurement that delivered a definitive TOC determination. CBL-VDL data offered no practical value for TOC placement in such extreme casing geometry and should not be relied upon as a primary cement evaluation method in similar conditions.

Discussion

The findings from this case study underscore the significant advancements in cement evaluation technologies, particularly for very large-diameter and thick-walled casings. Historically, both traditional CBL-VDL and standard ultrasonic tools have struggled to provide accurate cement assessment in such extreme well conditions due to signal attenuation, centralization challenges, and software limitations.

In this case, the CBL-VDL measurement did not provide a reliable indication of the TOC. The minimal amplitude contrast between free pipe and bonded cement, combined with the inherent limitations of sonic energy transmission through heavy-wall casing, rendered the CBL-VDL response inconclusive. Even with proper tool centralization, the amplitude values from free pipe and bonded cement were too close to allow for confident interpretation. This confirms that CBL-VDL should not be considered a viable primary cement evaluation method for conductor strings above 22 inches OD with wall thicknesses greater than 1 inch.

In contrast, the pulse-echo ultrasonic tool—equipped with a next-generation high-power transducer, large rotating logging head, and custom centralization—delivered clear, high-quality acoustic impedance data. This enabled precise identification of the TOC and accurate differentiation between cemented and uncemented sections.

Although the primary scope was cement evaluation, the high fidelity of the pulse-echo measurements also delivered accurate casing thickness and radius data. These measurements clearly resolved weld locations and showed excellent repeatability, suggesting they could be valuable for **future corrosion and mechanical integrity assessments** in similar conductor strings.

In addition, a correlation was observed between the casing thickness curve and acoustic impedance values. While the cause of this relationship is not fully understood, it is a secondary observation that may merit further investigation, particularly in thick-walled, large-diameter casings.

As operators continue to deploy larger and heavier conductor strings in complex offshore environments, the demand for cement evaluation solutions that can deliver definitive TOC placement in such conditions will grow. This case confirms that advanced low-frequency, high-power ultrasonic technology currently provides the only reliable solution for these extreme applications.

Conclusion

This case represents the first documented successful cement evaluation in a 26-inch OD, 1.25-inch wall conductor casing. The application of next-generation pulse-echo ultrasonic technology enabled a clear and definitive determination of the top of cement (TOC), whereas CBL-VDL measurements provided insufficient amplitude contrast for reliable interpretation in this casing size and wall thickness.

The combination of high-power transducers, large rotating logging heads, optimized centralization, and expanded processing algorithms extended the operating envelope of ultrasonic cement evaluation far beyond the limits of legacy tools. This capability was critical in confirming cement placement despite the absence of visual returns, allowing drilling operations to proceed safely and confidently without the need for a costly re-spud. The approach saved several days of rig time, reduced operational costs by avoiding additional conductor installation, and contributed to lower carbon emissions by eliminating unnecessary rework.

In addition to achieving the primary objective, the pulse-echo measurements delivered accurate casing thickness and radius data, even in such large-diameter, thick-walled casing. This capability offers potential for future applications in corrosion monitoring and mechanical integrity assessments, providing long-term value beyond the immediate cement evaluation.

Acknowledgements

The authors thank BP UK for granting permission to publish this work. Appreciation is extended to the operations and engineering teams for their invaluable contributions to data acquisition, interpretation, and field implementation. Special recognition is given to Abe Vereide (BP) for his technical input during pre-job and post-job discussions. The authors also acknowledge the support of colleagues and collaborators whose insights and technical expertise greatly contributed to this study.

Nomenclature

AI = Acoustic impedance
CBL = Cement bond log
VDL = Variable density log
OD = Casing outer diameter
ID = Casing internal diameter
TOC = Top of cement
ROV = Remotely operated vehicle
BOP = Blowout preventer

Symbols

Z = Acoustic impedance, MRayl
ρ = Density, g/cm^3
ν = Acoustic velocity, m/s

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